

Developer's setup instructions

How to Start

1. Set Up Local Development Environment

To begin, set up and activate a local (virtual) development environment to ensure all dependencies and settings are isolated. Follow these steps:

```
$ python -m venv venv
```

On macOS and Linux:

```
$ source venv/bin/activate
```

On Windows:

```
$ source venv/Scripts/activate
```

1.1. Install Python (3.13.1)

Make sure that you have Python 3.13.1 installed. You can download it from the official Python website: [Python Downloads](#).

1.2. Install Node.js (22.14.0) and Node Package Manager (npm) (10.9.2)

Install Node.js and npm by following the instructions on the official Node.js website: [Node.js Downloads](#).
On Windows:

```
$ winget install Schniz.fnm  
$ fnm install 22.14.0
```

On macOS and Linux:

```
$ curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.2/install.sh | bash  
$ nvm install 22.14.0
```

1.3. Activate Virtual Environment and Install Dependencies

```
$ python -m venv venv  
$ source venv/bin/activate  
$ pip install -r requirements.txt
```

2. Populate or Delete Database with Initial Data

To populate or delete the database with initial data:

```
$ python manage.py seed
```

To delete the database and re-seed it with initial data:

```
$ python manage.py unseed
```

3. Run Test

To ensure everything is working correctly, run all tests:

```
$ python manage.py test
```

4. Run the Django Development Server

To run the Django development server, use the following command:

```
$ python manage.py runserver
```

The server will be hosted at <http://127.0.0.1:8000/> by default.

If you want to specify a different port, you can use the following command:

```
$ python manage.py runserver <port_number>
```

5. Run the React Development Server

Before running the React development server, ensure that you have set the appropriate environment variables in the `.env` file. You can do this by creating a `.env` file in the root directory of the project and adding the following lines:

```
$ echo FRONTEND_BASE_URL=http://localhost:3000 >> .env
```

If you want to specify a different port, you can use the following command:

```
$ echo FRONTEND_BASE_URL=http://your_url_port >> .env
```

To run the React development server, use the following command:

```
$ npm run dev
```

Practical User Instances in Seed Data

The following user instances will only be created after running the command `$ python manage.py seed`:

- User: `@john_doe`

- A default-generated Manager account for the Book Club, intended for accessing Club Manager-specific functionalities.
- Default password: `Password123!`
- User: `@admin`
 - To ensure system security, only administrator accounts can register new administrator users. Therefore, an initial administrator account is necessary.
 - The `@admin` user is an administrator account explicitly created via `seed_user.py`, enabling access to functionalities that require admin permissions.
 - Default password: `Password123!`

Deployed Version

You can access the deployed version of this system at <http://51.21.191.188:8000/>.

Troubleshooting

Common Errors:

Database Connection Error: Ensure that the database credentials in `settings.py` are correctly set for your local or production environment.

Missing Environment Variables: If the application fails to start, double-check that all necessary environment variables (e.g., `DJANGO_SECRET_KEY`, `REACT_APP_API_URL`) are set correctly.

Frontend Build Errors: If there are issues with building the React app, try deleting the `node_modules` folder and running `npm install` again.